A2 Q4

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Example Suppose $\{a_j\}$ is a linear filter and suppose
$m_t = c_0 + c_1 t + \ldots + c_k t^k$ is a polynomial of degree k
Show that the filter passes m_t exactly ($m_t = \sum_{j=-\infty}^{\infty} a_j m_{t-j}$) iff:
i) $\sum_{j=-\infty}^{\infty} a_j = 1$ (1 condition)
ii) $\sum_{j=-\infty}^{\infty} j^r a_j = 0$ for $r = 1, \ldots, k$ (k conditions)
Not hard to check, but it is an iff so all $k+1$ conditions must be satisfied.

☆ **Solution:** The output of the filter $\{a_j\}$ is

$$\sum_{j=-\infty}^{\infty} a_j m_{t-j} = \sum_{j=-\infty}^{\infty} a_j (c_0 + c_1(t-j) + \ldots + c_k(t-j)^k)$$
$$= \sum_{j=-\infty}^{\infty} a_j \sum_{i=0}^{k} c_i (t-j)^i$$
$$= \sum_{j=-\infty}^{\infty} a_j c_0 + \sum_{j=-\infty}^{\infty} a_j \sum_{i=1}^{k} c_i (t-j)^i \quad \text{☆ Hint: Use Binomial Theorem}$$
$$= \sum_{j=-\infty}^{\infty} a_j c_0 + \sum_{j=-\infty}^{\infty} a_j \sum_{i=1}^{k} c_i \sum_{r=0}^{i} \binom{i}{r} (-j)^r t^{i-r} \quad \left(\binom{i}{r} = {}_iC_r\right)$$
$$= \sum_{j=-\infty}^{\infty} a_j c_0 + \underbrace{\sum_{j=-\infty}^{\infty} a_j \sum_{i=1}^{k} c_i t^i}_{r=0 \text{ terms}} + \sum_{j=-\infty}^{\infty} a_j \sum_{i=1}^{k} c_i \sum_{r=1}^{i} \binom{i}{r} (-j)^r t^{i-r}$$
(c: note, not 0 any more)
$$= m_t \sum_{j=-\infty}^{\infty} a_j + \sum_{i=1}^{k} \sum_{r=1}^{i} c_i \binom{i}{r} (-1)^r t^{i-r} \sum_{j=-\infty}^{\infty} j^r a_j$$

(Unfortunately) All of these infinite sum manipulations happen often in practice w/ time series. need to become adept at doing these — Prof. Glen Takahara

Want other terms = 0 ☆ Upper limit depends on middle sum (variable j)

Remainder of sol'n: Show that it all = m_t if sum. $a_j = 1$,
AND sum $j^r a_j = 0$
(can show this by showing all coefs. = 0)
Taylor Series expansion has similar argument

$$= m_t \sum_{-\infty}^{\infty} a_j + \sum_{r=1}^{k} \underbrace{\sum_{i=r}^{k} c_i \binom{i}{r} (-1)^r t^{i-r}}_{f_r(t) \text{ where}} \underbrace{\sum_{j=-\infty}^{\infty} j^r a_j}_{C_r}$$

$$f_r(t) = \sum_{i=r}^{k} c_i \binom{i}{r} (-1)^r t^{i-r}$$

and $C_r = \sum_{j=-\infty}^{\infty} j^r a_j$ for $r = 0, \ldots, k$

$$= m_t C_0 + \sum_{r=1}^{k} f_r(t) C_r$$

i) To show $\sum_{j=-\infty}^{\infty} a_j = 1$,
we use the last line of the prior equation and
see that $\sum_{j=-\infty}^{\infty} a_j = C_0 = \sum_{j=-\infty}^{\infty} j^0 a_j$

Assume ii) holds, i.e. $C_r = 0$ for $r \geq 1$. Then:

$$m_t = \sum_{j=-\infty}^{\infty} a_j m_{t-j}$$
$$= m_t \sum_{j=-\infty}^{\infty} a_j + \sum_{r=1}^{k} f_r(t) \cdot 0$$
$$= m_t C_0 \leftarrow C_0 \neq 0 \text{ b/c ii) only for } r \geq 1$$
$$m_t = m_t C_0$$

Clearly, $C_0 = \sum_{j=-\infty}^{\infty} a_j$ must $= 1$ in order for the equality to hold.
∴ i) holds

ii) To show $C_r = \sum_{j=-\infty}^{\infty} j^r a_j = 0 \ \forall r$, we must show the coe
☆ I.e., show C_r is causal (i.e. $a_j = 0$ when $j < 0$) and

ii) Now, we must show that $C_r = 0$ for $r \geq 1$.
Assume i) holds, i.e. $C_0 = 1$ for $r = 0$. Then:

$$m_t = \sum_{j=-\infty}^{\infty} a_j m_{t-j}$$
$$= m_t \sum_{j=-\infty}^{\infty} a_j + \sum_{r=1}^{k} f_r(t) C_r$$
$$= m_t C_0 + \sum_{r=1}^{k} \sum_{i=r}^{k} c_i \binom{i}{r} (-1)^r t^{i-r} C_r$$

Note that $f_r(t)$ is a polynomial f'n of t, with coefficients:
$c_i \binom{i}{r} C_r$
Since $\binom{i}{r}$ will always be a non-integer b/c $r \geq 1$ and

Since $\binom{i}{r}$ will always be a positive integer for $r \geq 1$ and $i = r$, and c_i cannot be assumed to be equal to 0 $\forall i$, then it follows that C_r must equal to 0 for the above equality to hold.

$$\begin{aligned} m_t &= m_t C_0 + \sum_{r=1}^{k} \sum_{i=r}^{k} c_i \binom{i}{r} C_r t^{i-r} \\ &= m_t(1) + \sum_{r=1}^{k} \sum_{i=r}^{k} c_i \binom{i}{r} (0)\, t^{i-r} \\ &= m_t + 0 \\ m_t &= m_t \end{aligned}$$

$\therefore$ ii) holds

$\therefore$ We have shown that $m_t = \sum_{j=-\infty}^{\infty} a_j m_{t-j} \Rightarrow C_0 = 1$ and $C_r = 0$ for $r \geq 1$

Now, to finalize, we show the inverse direction $C_0 = 1$ and $C_r = 0$ for $r \geq 1 \Rightarrow m_t = \sum_{j=-\infty}^{\infty} a_j m_{t-j}$

$$\begin{aligned} m_t &= \sum_{j=-\infty}^{\infty} a_j m_{t-j} \\ &= m_t \sum_{j=-\infty}^{\infty} a_j + \sum_{r=1}^{k} f_r(t) C_r \\ &= m_t C_0 + \sum_{r=1}^{k} f_r(t)(0) \\ &= m_t(1) + 0 \\ &= m_t \end{aligned}$$

$\therefore$ We have shown that $m_t = \sum_{j=-\infty}^{\infty} a_j m_{t-j}$ iff

$$\begin{cases} \sum_{j=-\infty}^{\infty} a_j = 1 & \text{and} \\ \sum_{j=-\infty}^{\infty} j^r a_j = 0 & \text{for } r = 1, \ldots, k \end{cases}$$

$\blacksquare$